MTH208: Worksheet 6

Web Scraping Basics

In this worksheet, we will learn the basics of scraping the web using **R**. Web scraping means collecting information from a website in a way that it can be stored and analyzed easily. We will use the package `rvest` for scraping, and the `tidyverse` for data handling.

```
library(rvest)
library(tidyverse)
```

The first step in scraping is to read the HTML of a webpage using the function `read_html()`. The code block given below extracts and prints the title of my homepage.

```
{r}
#| eval: false
url <- "https://home.iitk.ac.in/~akasha/index.html"
page <- read_html(url)

# Extract the page title
title <- page %>% html_element("title") %>% html_text()
print(title)
```

Now change the URL to https://www.wikipedia.org and print the page title.

The following code block demonstrates how to gather all hyperlinks from on a webpage.

```
{r}
#| eval: false
url <- "https://home.iitk.ac.in/~akasha/researh.html"
page <- read_html(url)

links <- page %>% html_elements("a") %>% html_attr("href")

head(links, 10)
```

How many total links are there on my research page?

Practice Problems

1. From https://home.iitk.ac.in/~akasha/research.html, extract all image URLs.

2. Try the following code block that extracts a structured table of CRAN mirror sites from the CRAN website and print the URLs of the first 5 mirror sites.

```
{r}
#| eval: false
url <- "https://cran.r-project.org/mirrors.html"
page <- read_html(url)

urls <- page %>% html_elements("td a") %>% html_attr("href")
head(urls)
```

   Write a code to extract the server names (not the URLs).

3. Scrape the first 5 quotes from https://quotes.toscrape.com and save them into a CSV file with two columns: `Quote` and `Author`.

4. Write an R code to obtain the following information about the best 100 movies on Netflix right now (and store the information in a clean data.frame) from the URL https://editorial.rottentomatoes.com/guide/best-netflix-movies-to-watch-right-now/.

   The website has the "100 best movies on Netflix" ranked by "Tomatometer". Scrape this website and collect the following information:

   1. Ranking
   2. Name of Movie
   3. Tomato % score
   4. Year of movie
   5. Director(s) of the movie

   You may not be able to get a clean list of the movie names. Using functions `strsplit`, `substring`, `gsub`, `substr` (or any other methods) create a final data frame of the movies. The data frame must be created like:

```
netflix_movies <- data.frame("Name" = name, "Year" = year, ...)
```

5. Using `html_table` scrape the table of Women's Tennis Association ranking of players below, and produce a clean table of Rank, Name, Points, and age.

   https://www.espn.com/tenis/rankings/_/tipo/wta